

12 june

In [8]:

```
import pandas as pd
```

In [10]:

```
pd.__version__
```

Out[10]:

```
'2.2.2'
```

In [12]:

```
#pip install --upgrade openpxyl
```

In [14]:

```
emp = pd.read_excel(r"C:\Users\kavya\OneDrive\Documents\12 raw data set.xlsx")
```

In [15]:

```
emp
```

Out[15]:

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience#\$	34 years	Mumbai	5^00#0	2+
1	Teddy^	Testing	45' yr	Bangalore	10%%000	<3
2	Uma#r	Dataanalyst^ ^#	NaN	NaN	1\$5%000	4> yrs
3	Jane	Ana^ ^lytics	NaN	Hyderbad	2000^0	NaN
4	Uttam*	Statistics	67-yr	NaN	30000-	5+ year
5	Kim	NLP	55yr	Delhi	6000^\$0	10+

In [18]:

```
id(emp)
```

Out[18]:

```
2302240290096
```

In [20]:

```
emp.columns
```

Out[20]:

```
Index(['Name', 'Domain', 'Age', 'Location', 'Salary', 'Exp'], dtype='object')
```

In [22]:

```
emp.shape
```

Out[22]:

```
(6, 6)
```

In [24]:

```
emp.shape
```

Out[24]:

```
(6, 6)
```

In [26]:

```
emp.head()
```

Out[26]:

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience#\$	34 years	Mumbai	5^00#0	2+
1	Teddy^	Testing	45' yr	Bangalore	10%%000	<3
2	Uma#r	Dataanalyst^ ^#	NaN	NaN	1\$5%000	4> yrs
3	Jane	Ana^ ^lytics	NaN	Hyderbad	2000^0	NaN
4	Uttam*	Statistics	67-yr	NaN	30000-	5+ year

In [28]:

```
emp.tail()
```

Out[28]:

	Name	Domain	Age	Location	Salary	Exp
1	Teddy^	Testing	45' yr	Bangalore	10%%000	<3
2	Uma#r	Dataanalyst^^#	NaN	NaN	1\$5%000	4> yrs
3	Jane	Ana^^lytics	NaN	Hyderabad	2000^0	NaN
4	Uttam*	Statistics	67-yr	NaN	30000-	5+ year
5	Kim	NLP	55yr	Delhi	6000^\$0	10+

In [30]:

emp.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6 entries, 0 to 5
Data columns (total 6 columns):
#   Column      Non-Null Count  Dtype
---  -
0    Name        6 non-null      object
1    Domain       6 non-null      object
2    Age          4 non-null      object
3    Location     4 non-null      object
4    Salary       6 non-null      object
5    Exp          5 non-null      object
dtypes: object(6)
memory usage: 420.0+ bytes
```

In [32]:

emp

Out[32]:

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience#\$	34 years	Mumbai	5^00#0	2+
1	Teddy^	Testing	45' yr	Bangalore	10%%000	<3
2	Uma#r	Dataanalyst^^#	NaN	NaN	1\$5%000	4> yrs
3	Jane	Ana^^lytics	NaN	Hyderabad	2000^0	NaN
4	Uttam*	Statistics	67-yr	NaN	30000-	5+ year
5	Kim	NLP	55yr	Delhi	6000^\$0	10+

In [34]:

emp.isnull()

Out[34]:

	Name	Domain	Age	Location	Salary	Exp
0	False	False	False	False	False	False
1	False	False	False	False	False	False
2	False	False	True	True	False	False
3	False	False	True	False	False	True
4	False	False	False	True	False	False
5	False	False	False	False	False	False

In [36]:

emp.isna()

Out[36]:

	Name	Domain	Age	Location	Salary	Exp
0	False	False	False	False	False	False
1	False	False	False	False	False	False
2	False	False	True	True	False	False
3	False	False	True	False	False	True
4	False	False	False	True	False	False
5	False	False	False	False	False	False

In [38]:

emp.isnull().sum()

Out[38]:

```

Name      0
Domain    0
Age       2
Location  2
Salary    0
Exp       1
dtype: int64

```

Data cleaning or data cleansimg

In [41]:

emp

Out[41]:

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience#\$	34 years	Mumbai	5^00#0	2+
1	Teddy^	Testing	45' yr	Bangalore	10%%000	<3
2	Uma#r	Dataanalyst^^#	NaN	NaN	1\$5%000	4> yrs
3	Jane	Ana^^lytics	NaN	Hyderbad	2000^0	NaN
4	Uttam*	Statistics	67-yr	NaN	30000-	5+ year
5	Kim	NLP	55yr	Delhi	6000^\$0	10+

In [43]:

emp['Name']

Out[43]:

```

0      Mike
1    Teddy^
2    Uma#r
3      Jane
4    Uttam*
5        Kim
Name: Name, dtype: object

```

In [45]:

emp['Name'] = emp['Name'].str.replace(r'\W', '', regex=True)

In [47]:

emp['Name']

Out[47]:

```

0      Mike
1    Teddy
2    Umar
3      Jane
4    Uttam
5        Kim
Name: Name, dtype: object

```

In [49]:

emp

Out[49]:

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience#\$	34 years	Mumbai	5^00#0	2+
1	Teddy	Testing	45' yr	Bangalore	10%%000	<3
2	Umar	Dataanalyst^^#	NaN	NaN	1\$5%000	4> yrs
3	Jane	Ana^^lytics	NaN	Hyderbad	2000^0	NaN
4	Uttam	Statistics	67-yr	NaN	30000-	5+ year
5	Kim	NLP	55yr	Delhi	6000^\$0	10+

In [51]:

emp['Domain']

Out[51]:

```

0      Datascience#$
1        Testing
2    Dataanalyst^^#
3      Ana^^lytics
4        Statistics
5          NLP
Name: Domain, dtype: object

```

In [53]:

```
emp['Domain']= emp['Domain'].str.replace(r'\W','',regex=True)
```

In [55]:

```
emp['Domain']
```

Out[55]:

```
0    Datascience
1      Testing
2    Dataanalyst
3      Analytics
4      Statistics
5         NLP
Name: Domain, dtype: object
```

In [57]:

```
emp
```

Out[57]:

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34 years	Mumbai	5^00#0	2+
1	Teddy	Testing	45' yr	Bangalore	10%%000	<3
2	Umar	Dataanalyst	NaN	NaN	1\$5%000	4> yrs
3	Jane	Analytics	NaN	Hyderbad	2000^0	NaN
4	Uttam	Statistics	67-yr	NaN	30000-	5+ year
5	Kim	NLP	55yr	Delhi	6000^\$0	10+

In [59]:

```
emp['Age'] = emp['Age'].str.replace(r'\W','',regex=True)
```

In [61]:

```
emp['Age']
```

Out[61]:

```
0    34years
1     45yr
2     NaN
3     NaN
4     67yr
5     55yr
Name: Age, dtype: object
```

In [63]:

```
emp['Age'] = emp['Age'].str.extract('(\d+)')
```

```
<>:1: SyntaxWarning: invalid escape sequence '\d'
<>:1: SyntaxWarning: invalid escape sequence '\d'
C:\Users\kavya\AppData\Local\Temp\ipykernel_24384\1884116463.py:1: SyntaxWarning: invalid escape sequence '\d'
emp['Age'] = emp['Age'].str.extract('(\d+)')
```

In [65]:

```
emp['Age']
```

Out[65]:

```
0    34
1    45
2    NaN
3    NaN
4    67
5    55
Name: Age, dtype: object
```

In [67]:

```
emp
```

Out[67]:

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34	Mumbai	5^00#0	2+
1	Teddy	Testing	45	Bangalore	10%%000	<3
2	Umar	Dataanalyst	NaN	NaN	1\$5%000	4> yrs
3	Jane	Analytics	NaN	Hyderbad	2000^0	NaN
4	Uttam	Statistics	67	NaN	30000-	5+ year
5	Kim	NLP	55	Delhi	6000^\$0	10+

In [69]:

```
emp['Location'] = emp['Location'].str.replace(r'\W','',regex=True)
```

In [71]:

```
emp['Location']
```

Out[71]:

```
0      Mumbai
1    Bangalore
2         NaN
3    Hyderabad
4         NaN
5        Delhi
Name: Location, dtype: object
```

In [73]:

```
emp
```

Out[73]:

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34	Mumbai	5^00#0	2+
1	Teddy	Testing	45	Bangalore	10%%000	<3
2	Umar	Dataanalyst	NaN	NaN	1\$5%000	4> yrs
3	Jane	Analytics	NaN	Hyderabad	2000^0	NaN
4	Uttam	Statistics	67	NaN	30000-	5+ year
5	Kim	NLP	55	Delhi	6000^\$0	10+

In [75]:

```
emp['Salary'] = emp['Salary'].str.replace(r'\W','',regex=True)
```

In [77]:

```
emp['Salary']
```

Out[77]:

```
0      5000
1     10000
2     15000
3     20000
4     30000
5     60000
Name: Salary, dtype: object
```

In [79]:

```
emp.head()
```

Out[79]:

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34	Mumbai	5000	2+
1	Teddy	Testing	45	Bangalore	10000	<3
2	Umar	Dataanalyst	NaN	NaN	15000	4> yrs
3	Jane	Analytics	NaN	Hyderabad	20000	NaN
4	Uttam	Statistics	67	NaN	30000	5+ year

In [81]:

```
emp['Exp'] = emp['Exp'].str.extract('(\d+)')
```

```
<>:1: SyntaxWarning: invalid escape sequence '\d'
<>:1: SyntaxWarning: invalid escape sequence '\d'
C:\Users\kavya\AppData\Local\Temp\ipykernel_24384\3836251810.py:1: SyntaxWarning: invalid escape sequence '\d'
emp['Exp'] = emp['Exp'].str.extract('(\d+)')
```

In [83]:

```
emp['Exp']
```

Out[83]:

```
0      2
1      3
2      4
3     NaN
4      5
5     10
Name: Exp, dtype: object
```

In [85]:

```
emp
```

Out[85]:

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34	Mumbai	5000	2
1	Teddy	Testing	45	Bangalore	10000	3
2	Umar	Dataanalyst	NaN	NaN	15000	4
3	Jane	Analytics	NaN	Hyderbad	20000	NaN
4	Uttam	Statistics	67	NaN	30000	5
5	Kim	NLP	55	Delhi	60000	10

In [87]:

```
clean_data = emp.copy()
```

In [89]:

```
clean_data
```

Out[89]:

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34	Mumbai	5000	2
1	Teddy	Testing	45	Bangalore	10000	3
2	Umar	Dataanalyst	NaN	NaN	15000	4
3	Jane	Analytics	NaN	Hyderbad	20000	NaN
4	Uttam	Statistics	67	NaN	30000	5
5	Kim	NLP	55	Delhi	60000	10

EDA techniques ets apply

missing values treatement for the numerical data

In [93]:

```
clean_data
```

Out[93]:

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34	Mumbai	5000	2
1	Teddy	Testing	45	Bangalore	10000	3
2	Umar	Dataanalyst	NaN	NaN	15000	4
3	Jane	Analytics	NaN	Hyderbad	20000	NaN
4	Uttam	Statistics	67	NaN	30000	5
5	Kim	NLP	55	Delhi	60000	10

In [95]:

```
clean_data.isnull().sum()
```

Out[95]:

```
Name      0
Domain    0
Age       2
Location  2
Salary    0
Exp       1
dtype: int64
```

In [107]:

```
clean_data['Age']
```

Out[107]:

```
0    34
1    45
2    NaN
3    NaN
4    67
5    55
Name: Age, dtype: object
```

In [109]:

```
import numpy as np
```

In [111]:

```
clean_data['Age'] = clean_data['Age'].fillna(np.mean(pd.to_numeric(clean_data['Age'])))
```

In [117]:

```
clean_data['Age']
```

Out[117]:

```
0      34
1      45
2    50.25
3    50.25
4      67
5      55
Name: Age, dtype: object
```

In [115]:

```
clean_data['Exp']
```

Out[115]:

```
0      2
1      3
2      4
3    NaN
4      5
5     10
Name: Exp, dtype: object
```

In [119]:

```
clean_data['Exp'] = clean_data['Exp'].fillna(np.mean(pd.to_numeric(clean_data['Exp'])))
```

In [121]:

```
clean_data['Exp']
```

Out[121]:

```
0      2
1      3
2      4
3    4.8
4      5
5     10
Name: Exp, dtype: object
```

In [123]:

```
clean_data
```

Out[123]:

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34	Mumbai	5000	2
1	Teddy	Testing	45	Bangalore	10000	3
2	Umar	Dataanalyst	50.25	NaN	15000	4
3	Jane	Analytics	50.25	Hyderbad	20000	4.8
4	Uttam	Statistics	67	NaN	30000	5
5	Kim	NLP	55	Delhi	60000	10

In [127]:

```
clean_data['Location'].isnull().sum()
```

Out[127]:

```
2
```

In [129]:

```
clean_data['Location'] = clean_data['Location'].fillna(clean_data['Location'].mode()[0])
```

In [131]:

```
clean_data['Location']
```

Out[131]:

```
0      Mumbai
1    Bangalore
2    Bangalore
3    Hyderbad
4    Bangalore
5        Delhi
Name: Location, dtype: object
```

In [133]:

```
clean_data
```

Out[133]:

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34	Mumbai	5000	2
1	Teddy	Testing	45	Bangalore	10000	3
2	Umar	Dataanalyst	50.25	Bangalore	15000	4
3	Jane	Analytics	50.25	Hyderbad	20000	4.8
4	Uttam	Statistics	67	Bangalore	30000	5
5	Kim	NLP	55	Delhi	60000	10

In [135]:

emp.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6 entries, 0 to 5
Data columns (total 6 columns):
#   Column      Non-Null Count  Dtype
---  ---
0    Name        6 non-null      object
1    Domain       6 non-null      object
2    Age          4 non-null      object
3    Location     4 non-null      object
4    Salary       6 non-null      object
5    Exp          5 non-null      object
dtypes: object(6)
memory usage: 420.0+ bytes
```

In [137]:

clean_data.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6 entries, 0 to 5
Data columns (total 6 columns):
#   Column      Non-Null Count  Dtype
---  ---
0    Name        6 non-null      object
1    Domain       6 non-null      object
2    Age          6 non-null      object
3    Location     6 non-null      object
4    Salary       6 non-null      object
5    Exp          6 non-null      object
dtypes: object(6)
memory usage: 420.0+ bytes
```

In [139]:

clean_data['Age'] = clean_data['Age'].astype(int)

In [141]:

clean_data.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6 entries, 0 to 5
Data columns (total 6 columns):
#   Column      Non-Null Count  Dtype
---  ---
0    Name        6 non-null      object
1    Domain       6 non-null      object
2    Age          6 non-null      int32
3    Location     6 non-null      object
4    Salary       6 non-null      object
5    Exp          6 non-null      object
dtypes: int32(1), object(5)
memory usage: 396.0+ bytes
```

In [143]:

clean_data['Salary'] = clean_data['Salary'].astype(int)

In [145]:

clean_data['Exp'] = clean_data['Exp'].astype(int)

In [147]:

clean_data.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6 entries, 0 to 5
Data columns (total 6 columns):
#   Column      Non-Null Count  Dtype
---  ---
0    Name        6 non-null      object
1    Domain       6 non-null      object
2    Age          6 non-null      int32
3    Location     6 non-null      object
4    Salary       6 non-null      int32
5    Exp          6 non-null      int32
dtypes: int32(3), object(3)
memory usage: 348.0+ bytes
```

In [149]:


```
clean_data['Name'] = clean_data['Name'].astype('category')
clean_data['Doamin'] = clean_data['Domain'].astype('category')
clean_data['Location'] = clean_data['Location'].astype('category')
```

In [151]:

```
clean_data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6 entries, 0 to 5
Data columns (total 7 columns):
#   Column      Non-Null Count  Dtype
---  ---
0   Name        6 non-null      category
1   Domain      6 non-null      object
2   Age         6 non-null      int32
3   Location    6 non-null      category
4   Salary      6 non-null      int32
5   Exp         6 non-null      int32
6   Doamin      6 non-null      category
dtypes: category(3), int32(3), object(1)
memory usage: 914.0+ bytes
```

In [153]:

```
clean_data
```

Out[153]:

	Name	Domain	Age	Location	Salary	Exp	Doamin
0	Mike	Datascience	34	Mumbai	5000	2	Datascience
1	Teddy	Testing	45	Bangalore	10000	3	Testing
2	Umar	Dataanalyst	50	Bangalore	15000	4	Dataanalyst
3	Jane	Analytics	50	Hyderbad	20000	4	Analytics
4	Uttam	Statistics	67	Bangalore	30000	5	Statistics
5	Kim	NLP	55	Delhi	60000	10	NLP

In [155]:

```
clean_data.to_csv('clean_data.csv')
```

In [157]:

```
import os
os.getcwd()
```

Out[157]:

```
'C:\\Users\\kavya'
```

In [159]:

```
clean_data
```

Out[159]:

	Name	Domain	Age	Location	Salary	Exp	Doamin
0	Mike	Datascience	34	Mumbai	5000	2	Datascience
1	Teddy	Testing	45	Bangalore	10000	3	Testing
2	Umar	Dataanalyst	50	Bangalore	15000	4	Dataanalyst
3	Jane	Analytics	50	Hyderbad	20000	4	Analytics
4	Uttam	Statistics	67	Bangalore	30000	5	Statistics
5	Kim	NLP	55	Delhi	60000	10	NLP

In [161]:

```
import matplotlib.pyplot as plt
import seaborn as sns
```

In [163]:

```
import warnings
warnings.filterwarnings('ignore')
```

In [165]:

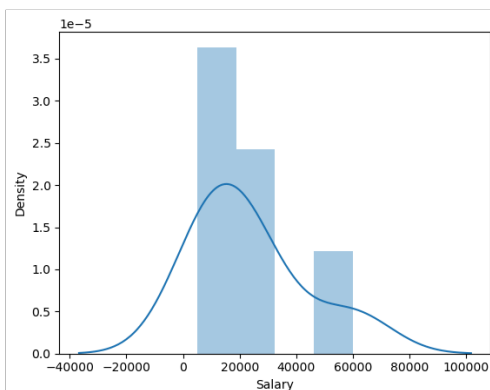
```
clean_data['Salary']
```

Out[165]:

```
0    5000
1   10000
2   15000
3   20000
4   30000
5   60000
Name: Salary, dtype: int32
```

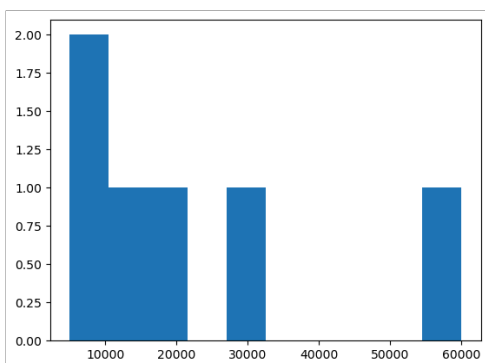
In [169]:

```
vis1 = sns.distplot(clean_data['Salary'])
```



In [171]:

```
vis2 = plt.hist(clean_data['Salary'])
```



In [173]:

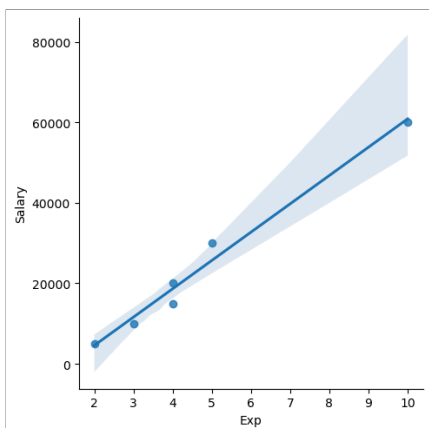
```
clean_data
```

Out[173]:

	Name	Domain	Age	Location	Salary	Exp	Doamin
0	Mike	Datascience	34	Mumbai	5000	2	Datascience
1	Teddy	Testing	45	Bangalore	10000	3	Testing
2	Umar	Dataanalyst	50	Bangalore	15000	4	Dataanalyst
3	Jane	Analytics	50	Hyderbad	20000	4	Analytics
4	Uttam	Statistics	67	Bangalore	30000	5	Statistics
5	Kim	NLP	55	Delhi	60000	10	NLP

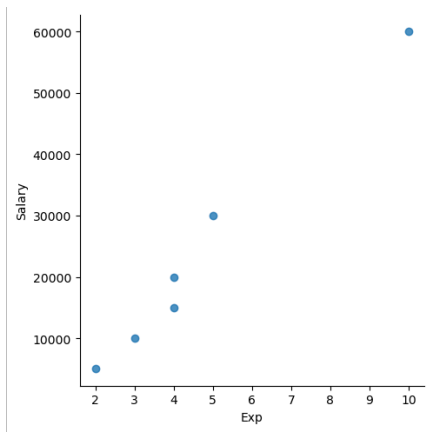
In [177]:

```
vis4 = sns.lmplot(data=clean_data, x = 'Exp', y='Salary')
```



In [179]:

```
vis4 = sns.lmplot(data=clean_data, x = 'Exp', y='Salary', fit_reg = False)
```



In [181]:

clean_data

Out[181]:

	Name	Domain	Age	Location	Salary	Exp	Doamin
0	Mike	Datascience	34	Mumbai	5000	2	Datascience
1	Teddy	Testing	45	Bangalore	10000	3	Testing
2	Umar	Dataanalyst	50	Bangalore	15000	4	Dataanalyst
3	Jane	Analytics	50	Hyderbad	20000	4	Analytics
4	Uttam	Statistics	67	Bangalore	30000	5	Statistics
5	Kim	NLP	55	Delhi	60000	10	NLP

In [183]:

X_iv = clean_data[['Name', 'Domain', 'Age', 'Location', 'Exp']]

In [185]:

X_iv

Out[185]:

	Name	Domain	Age	Location	Exp
0	Mike	Datascience	34	Mumbai	2
1	Teddy	Testing	45	Bangalore	3
2	Umar	Dataanalyst	50	Bangalore	4
3	Jane	Analytics	50	Hyderbad	4
4	Uttam	Statistics	67	Bangalore	5
5	Kim	NLP	55	Delhi	10

In [187]:

y_dv = clean_data[['Salary']]

In [189]:

y_dv

Out[189]:

	Salary
0	5000
1	10000
2	15000
3	20000
4	30000
5	60000

In [191]:

emp

Out[191]:

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34	Mumbai	5000	2
1	Teddy	Testing	45	Bangalore	10000	3
2	Umar	Dataanalyst	NaN	NaN	15000	4
3	Jane	Analytics	NaN	Hyderbad	20000	NaN
4	Uttam	Statistics	67	NaN	30000	5
5	Kim	NLP	55	Delhi	60000	10

In [193]:

clean_data

Out[193]:

	Name	Domain	Age	Location	Salary	Exp	Doamin
0	Mike	Datascience	34	Mumbai	5000	2	Datascience
1	Teddy	Testing	45	Bangalore	10000	3	Testing
2	Umar	Dataanalyst	50	Bangalore	15000	4	Dataanalyst
3	Jane	Analytics	50	Hyderbad	20000	4	Analytics
4	Uttam	Statistics	67	Bangalore	30000	5	Statistics
5	Kim	NLP	55	Delhi	60000	10	NLP

In [197]:

imputation = pd.get_dummies(clean_data, dtype=int)

In [199]:

imputation

Out[199]:

	Age	Salary	Exp	Name_Jane	Name_Kim	Name_Mike	Name_Teddy	Name_Umar	Name_Uttam	Domain_Analytics	...	Location_Bangalore	Locatic
0	34	5000	2	0	0	1	0	0	0	0	...	0	0
1	45	10000	3	0	0	0	1	0	0	0	...	1	1
2	50	15000	4	0	0	0	0	1	0	0	...	1	1
3	50	20000	4	1	0	0	0	0	0	1	...	0	0
4	67	30000	5	0	0	0	0	0	1	0	...	1	1
5	55	60000	10	0	1	0	0	0	0	0	...	0	0

6 rows × 25 columns

In [201]:

clean_data

Out[201]:

	Name	Domain	Age	Location	Salary	Exp	Doamin
0	Mike	Datascience	34	Mumbai	5000	2	Datascience
1	Teddy	Testing	45	Bangalore	10000	3	Testing
2	Umar	Dataanalyst	50	Bangalore	15000	4	Dataanalyst
3	Jane	Analytics	50	Hyderbad	20000	4	Analytics
4	Uttam	Statistics	67	Bangalore	30000	5	Statistics
5	Kim	NLP	55	Delhi	60000	10	NLP

In [203]:

len(clean_data)

Out[203]:

6

In [205]:

imputation.columns

Out[205]:

```
Index(['Age', 'Salary', 'Exp', 'Name_Jane', 'Name_Kim', 'Name_Mike',  
      'Name_Teddy', 'Name_Umar', 'Name_Uttam', 'Domain_Analytics',  
      'Domain_Dataanalyst', 'Domain_Datascience', 'Domain_NLP',  
      'Domain_Statistics', 'Domain_Testing', 'Location_Bangalore',  
      'Location_Delhi', 'Location_Hyderabad', 'Location_Mumbai',  
      'Doamin_Analytics', 'Doamin_Dataanalyst', 'Doamin_Datascience',  
      'Doamin_NLP', 'Doamin_Statistics', 'Doamin_Testing'],  
      dtype='object')
```

In [207]:

```
len(imputation.columns)
```

Out[207]:

25

In []: